# Exit Ticket · reflect and retrieve

## Exit Ticket

A rising scatter trend proves that extra practice alone caused higher marks.

## Use key vocabulary from memory

1. Explain data mining in your own words.

2. Explain scatter plot in your own words.

3. Explain outlier in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.



RHS COMPUTING / OCR H446 / LESSON 069 DRAFT · TEACHER REVIEW

# Do Now · retrieval

## Lesson 069 · Data Mining and Visualisation of Patterns

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

What is a backtracking branch?

### Q2

What is an outlier?

### Q3

What is the purpose of a scatter graph?

### Q4

Array [2,3,4,5]: double elements at indexes 1 and 3. Give updated array and sum.

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Define and predict — Data mining seeks patterns or associations across sets of data; define the question, prepare records and examine the results.

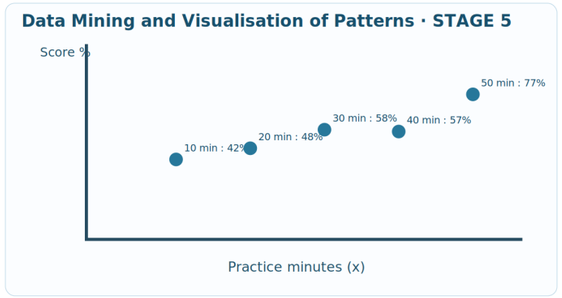
## Core idea 2

Explain — Visualisation makes patterns and anomalies visible. Scatter charts suit pairs of numeric data; bar charts compare categories.

## Core idea 3

Evaluate — A correlation does not establish cause. Sampling, confounders, missing values and measurement choices may mislead interpretation.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Evaluate mining a school's student-usage logs to predict attainment. Address fairness, data quality and privacy. [6]

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

For points (1,2),(2,4),(3,6), choose a suitable chart and describe association. [4]

## Question 2

For [40,42,46,96] calculate mean and median with working. [6]

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

data mining — Finding patterns in datasets.

scatter plot — Pairs of numeric variables plotted as points.

outlier — Unusually distant observation.

correlation — Association, not proof of causation.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

For points (1,2),(2,4),(3,6), state trend and one limitation.

## Follow the worked method

1. Follow the reasoning — Separate observation from causal explanation.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Follow the reasoning
The value96 pulls mean away from three clustered lower scores.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.